8 Scalar-Valued Functions of Several Variables

Definition 8.1: Scalar Function

A scalar-valued function of several variables maps a vector of multiple inputs to a single real output $f : D \subseteq \mathbb{R}^n \to \mathbb{R}$.

Examples:

- Temperature distribution: $T(\text{longitude}, \text{latitude}, \text{time}) : \mathbb{R}^3 \to \mathbb{R}$.
- Precipitation depth: $P(\text{longitude}, \text{latitude}) : \mathbb{R}^2 \to \mathbb{R}$.
- Domain and Range example: For $f(x, y) = y\sqrt{x-1}$,

$$\text{Domain: } \{(x, y) \in \mathbb{R}^2 : x \geq 1\}, \quad \text{Range: } \mathbb{R}$$

8.1 Visualizing Multivariable Functions: Traces and Level Sets

The graph of a function $f(x, y)$ is defined as the surface $\{(x, y, z) \in \mathbb{R}^3 : z = f(x, y)\}$.

Example 8.1 (Analyzing Traces of $f(x, y) = y \sin x$)**.**

To visualize $f(x, y) = y \sin x$, we examine slices along constant coordinates (traces):

- **y-traces ($y = c$):**

  For $y = 0 \implies z = 0$;

  for $y = 1 \implies z = \sin x$;

  for $y = -1 \implies z = -\sin x$. (Sinusoidal curves scaled by y).

- **x-traces ($x = c$):**

  For $x = 0 \implies z = 0$;

  for $x = \frac{\pi}{2} \implies z = y$;

  for $x = \pi \implies z = 0$. (Lines whose slopes depend on $\sin x$).

8.2 Level Sets and Contour Maps

Definition 8.2: Level Set

For a function $f : D \to \mathbb{R}$ ($D \subseteq \mathbb{R}^2$), the **level set** (or level curve) at height c is defined as

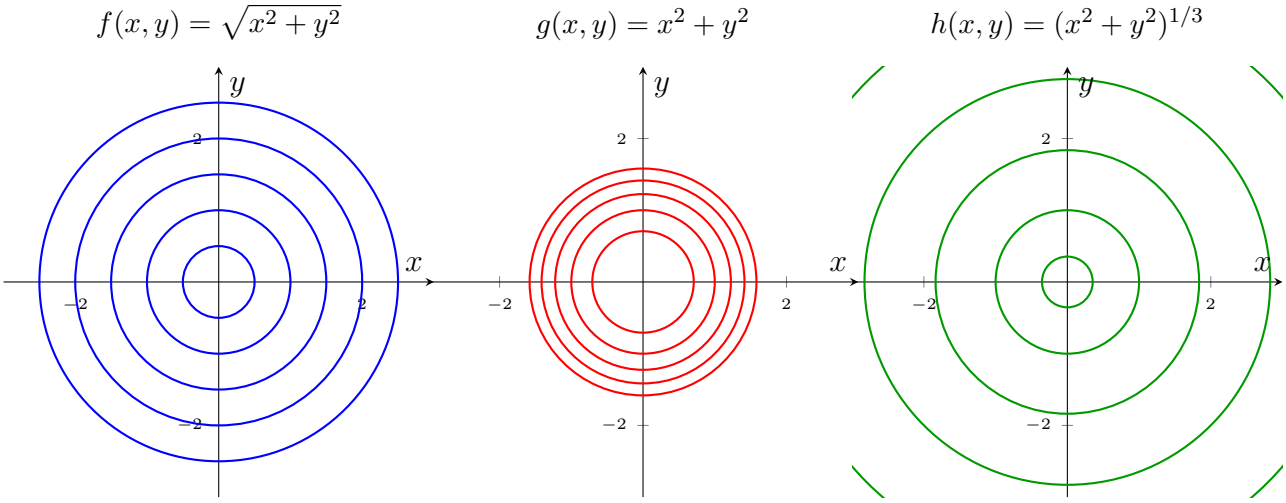
$$\{(x, y) \in \mathbb{R}^2 : f(x, y) = c\}.$$

A collection of such level curves forms a **contour map**.

Example 8.2 (Comparing Level Curves)**.**

Consider three functions and their level sets $f(x, y) = c$:

1. $f(x, y) = \sqrt{x^2 + y^2}$: Level sets are equally spaced concentric circles $x^2 + y^2 = c^2$ (a cone).
2. $g(x, y) = x^2 + y^2$: Level sets are circles $x^2 + y^2 = c$ spaced closer together as c increases (a paraboloid).
3. $h(x, y) = (x^2 + y^2)^{1/3}$: Level sets are circles $x^2 + y^2 = c^3$ spaced further apart as c increases.



8.3 Visualizing Higher Dimensions via Slices

- A sphere in $\mathbb{R}^3$ defined by $x^2 + y^2 + z^2 = \rho^2$ can be sliced along $z = c$ to produce 2D circular cross-sections in $\mathbb{R}^2$.

- A 3-sphere in $\mathbb{R}^4$ defined by $x^2 + y^2 + z^2 + w^2 = R^2$ can be visualized through its 3D cross-sections ($w = c$), which appear as concentric 2-spheres in $\mathbb{R}^3$ whose radii expand from 0 to R and then contract back to 0.

3D Sphere Slices ($z = c$) 4D Hypersphere Slices ($w = c$ in $\mathbb{R}^3$)



8.4 Limits of Functions of Several Variables

For single-variable functions, we say that the limit of a function $f(x)$ exists such that

$$\lim_{x \to a} f(x) = L \iff \forall \epsilon > 0, \exists \delta > 0 \text{ such that } 0 < |x - a| < \delta \implies |f(x) - L| < \epsilon$$

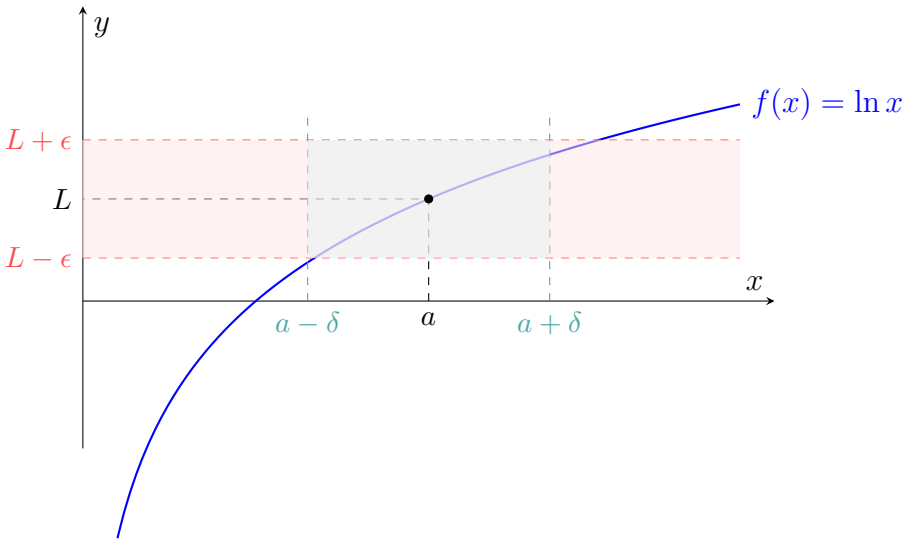
As shown below, we can visualize the ϵ-δ definition of a limit for a single-variable function $f(x) = \ln x$ at $x = a$.

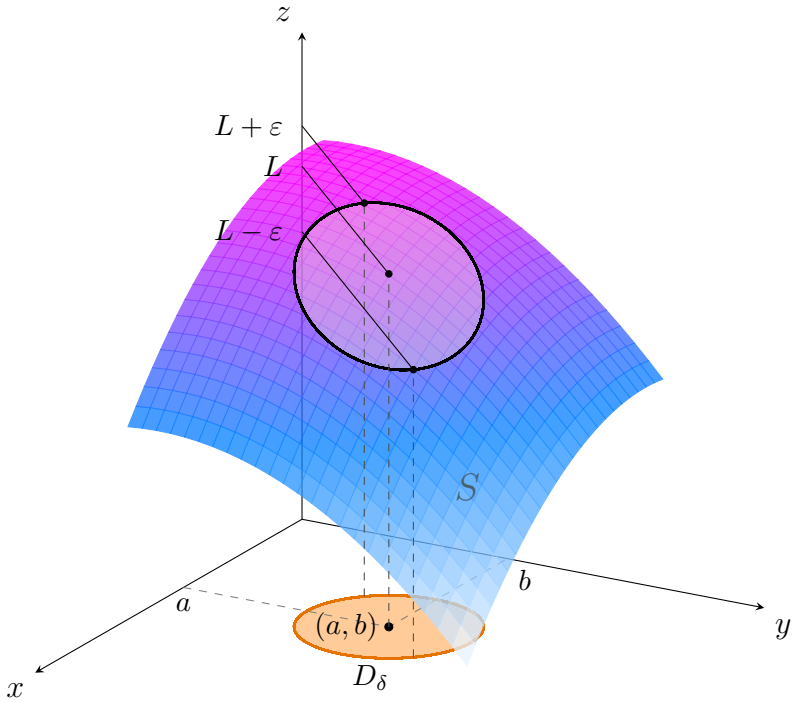
Now consider the case of functions of two variables. We say that the limit of a function $f(x, y)$ exists such that

$$\lim_{(x,y) \to (a,b)} f(x, y) = L \iff \forall \epsilon > 0, \exists \delta > 0 \text{ such that } 0 < \sqrt{(x - a)^2 + (y - b)^2} < \delta$$
$$\implies |f(x, y) - L| < \epsilon$$

Similarly, we can visualize the ϵ-δ definition of a limit for a two-variable function $f(x, y)$.

ϵ-δ Definition of Limit for $f(x) = \ln x$ at $x = a$.





Problem 8.1: Limit of a Function of Two Variables

Find, if it exists, $\lim_{(x,y)\to(0,0)} \dfrac{xy}{x^2+y^2}$.

Solution 8.1

First, we consider approaching $(0,0)$ along the x-axis, i.e., $y=0$:

$$\lim_{x\to 0} \frac{x(0)}{x^2+0^2} = 0$$

Next, we consider approaching $(0,0)$ along the y-axis, i.e., $x=0$:

$$\lim_{y\to 0} \frac{0(y)}{0^2+y^2} = 0$$

Next, we consider approaching $(0,0)$ along the line $y=x$:

$$\lim_{x\to 0} \frac{x(x)}{x^2+x^2} = \lim_{x\to 0} \frac{x^2}{2x^2} = \lim_{x\to 0} \frac{1}{2} = \frac{1}{2}$$

Since approaching along different paths yields different limiting values, the limit does not exist.

Systematic Path Approaches Checking paths individually can be cumbersome. Instead, we can systematically analyze general path families:

1. Linear Paths ($y=mx$):

$$\lim_{x\to 0} f(x,mx) = \lim_{x\to 0} \frac{x(mx)}{x^2+(mx)^2} = \lim_{x\to 0} \frac{mx^2}{x^2(1+m^2)} = \frac{m}{1+m^2}$$

The limit depends explicitly on the slope m. Hence, the limit does not exist.

2. Polar Coordinates ($x=r\cos\theta, y=r\sin\theta$):

$$\lim_{r\to 0} f(r\cos\theta, r\sin\theta) = \lim_{r\to 0} \frac{r^2\cos\theta\sin\theta}{r^2} = \cos\theta\sin\theta$$

The limit depends on the angle θ, confirming the limit does not exist.

Definition 8.3: Squeeze Theorem for Two Variables

If $g(x,y) \leq f(x,y) \leq h(x,y)$ for all (x,y) in a disk centered at (a,b) (except possibly at (a,b) itself), and

$$\lim_{(x,y)\to(a,b)} g(x,y) = \lim_{(x,y)\to(a,b)} h(x,y) = L$$

then $\lim_{(x,y)\to(a,b)} f(x,y) = L$.

Definition 8.4: Continuity of Functions of Several Variables

A function $f(x,y)$ is *continuous* at (a,b) if:

$$\lim_{(x,y)\to(a,b)} f(x,y) = f(a,b)$$

Furthermore, $f(x,y)$ is continuous on a domain D if it is continuous at every point $(a,b) \in D$.

Example 8.3 (Continuity of a Composite Function)**.**
Consider $f(x,y) = \arctan\left(\dfrac{y}{x}\right)$.

The inner function $g(x,y) = \dfrac{y}{x}$ is continuous everywhere except on the line $x = 0$. Since $h(t) = \arctan(t)$ is continuous on $\mathbb{R}$, the composite function $f(x,y) = h(g(x,y))$ is continuous at all points $(x,y) \in \mathbb{R}^2$ where $x \neq 0$.